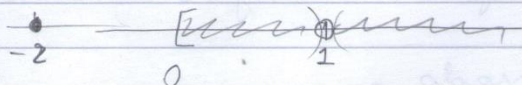


TP 2 – NOCIONES DE TOPOLOGÍA, FUNCIONES

1) Dom f $\left\{ \begin{array}{l} x(x+2)^4 \geq 0 \rightarrow x \geq 0 \vee x = -2 \\ x-1 \neq 0 \rightarrow x \neq 1 \end{array} \right.$



$P_1 = -2$ aislado - frontera
 $P_2 = 0$ frontera - de acumulación
 $P_3 = 2$ interior - de acumulación
 $P_4 = 1$ frontera - acumulación
 $P_5 = -3$ exterior

02) a) $\partial S = \text{Front } S = \{ (x,y) \in \mathbb{R}^2 / x^2 + y^2 = 4 \vee (x,y) = (0,0) \vee (x,y) = (2,2) \}$

b) $\overset{\circ}{S} = \text{Int } S = \{ (x,y) \in \mathbb{R}^2 / 0 < x^2 + y^2 < 4 \}$

c) $S' = \text{Ann } S = \{ (x,y) \in \mathbb{R}^2 / x^2 + y^2 \leq 4 \}$

d) S no es abierto (no todos sus pts son interiores, p.ej A)

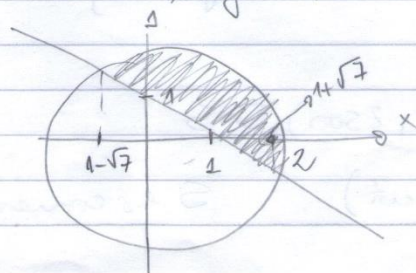
S no es cerrado ($\bar{0} \in S'$ pero $\bar{0} \notin S$)

S es acotado $\rightarrow \exists E(\bar{0}, 250) \supset S$

e) $\text{cl}(S) = S \cup S' = \{ (x,y) \in \mathbb{R}^2 / x^2 + y^2 \leq 4 \vee (x,y) = (2,2) \}$

3) a) $x^2 + y^2 - 4 \leq 0 \rightarrow x^2 + y^2 \leq 4$

$x + y \geq 1 \rightarrow x + y \geq 1 \rightarrow y \geq 1 - x$



$y = 1 - x \rightarrow x^2 + (1 - x)^2 = 4 \Rightarrow 0$

$\Rightarrow x^2 + 1 + x^2 - 2x = 4 \Rightarrow 0$

$\Rightarrow 2x^2 - 2x - 3 = 0$

$x_{1,2} = \frac{2 \pm \sqrt{4 + 24}}{4} = \frac{2 \pm \sqrt{28}}{2} = 1 \pm \sqrt{7}$

$$S = \{(x, y) \in \mathbb{R}^2 / x^2 + y^2 < 4 \wedge x + y > 1\}$$

$$\partial S = \{(x, y) \in \mathbb{R}^2 / (x + y = 1 \wedge 1 - \sqrt{7} \leq x \leq 1 + \sqrt{7}) \vee (x^2 + y^2 = 4 \wedge 1 - \sqrt{7} \leq x \leq 1 + \sqrt{7})\}$$

$$\text{Ext } S = \{(x, y) \in \mathbb{R}^2 / x^2 + y^2 > 4 \vee x + y < 1\}$$

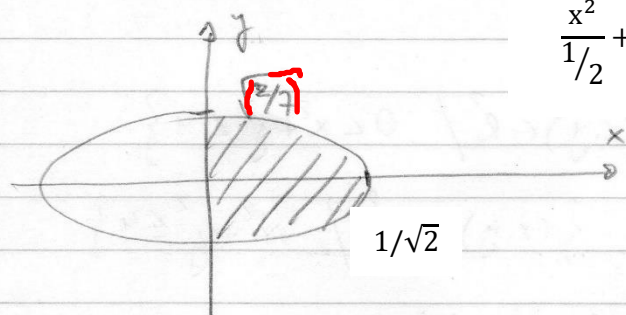
$$S' = S \rightarrow S \text{ es cerrado}$$

$$S \text{ es acotado} \rightarrow \exists E(\vec{0}, 2500) \supset S$$

S es compacto (cerrado y acotado)

S es convexo

$$3) b) \quad 4x^2 + 7y^2 \leq 2 \rightarrow 2x^2 + \frac{y^2}{2/7} \leq 1 \quad \wedge \quad x \geq 0$$



$$\frac{x^2}{1/2} + \frac{y^2}{2/7} \leq 1, \quad x \geq 0$$

$$S^\circ = \{ (x, y) \in \mathbb{R}^2 \mid 4x^2 + 7y^2 < 2 \wedge x > 0 \}$$

$$\partial S = \{ (x, y) \in \mathbb{R}^2 \mid (4x^2 + 7y^2 = 2 \wedge x \geq 0) \vee (x = 0 \wedge -\sqrt{\frac{2}{7}} \leq y \leq \sqrt{\frac{2}{7}}) \}$$

$$\text{Ext } S = \{ (x, y) \in \mathbb{R}^2 \mid x < 0 \vee 4x^2 + 7y^2 > 2 \}$$

$$S' = S \rightarrow S \text{ es cerrado}$$

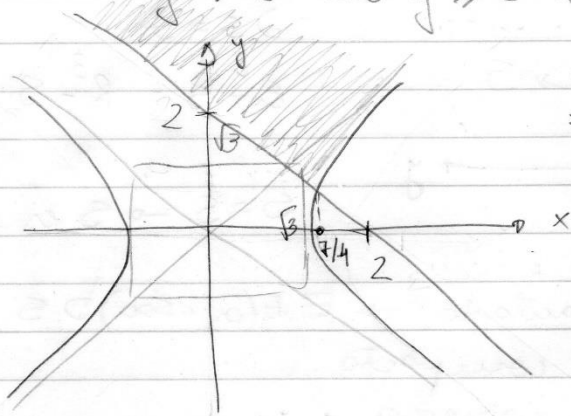
$$S \text{ acotado} \rightarrow \exists \epsilon \in (0, 2500) \supset S$$

S compacto (cerrado y acot)

S es conexo

$$3)c) \quad x^2 - y^2 \leq 3 \Rightarrow \frac{x^2}{3} - \frac{y^2}{3} \leq 1$$

$$x + y \geq 2 \Rightarrow y \geq 2 - x$$



$$\begin{cases} y = 2 - x \\ x^2 - (2 - x)^2 = 3 \end{cases} \Rightarrow 0$$

$$\Rightarrow x^2 - 4 + 4x = 3$$

$$4x = 7$$

$$x = 7/4$$

$$S = \{(x, y) \in \mathbb{R}^2 \mid x^2 - y^2 < 3 \wedge x + y > 2\}$$

$$\partial S = \{(x, y) \in \mathbb{R}^2 \mid (x + y = 2 \wedge x \leq 7/4) \vee (x^2 - y^2 = 3 \wedge x > 7/4 \wedge y > 0)\}$$

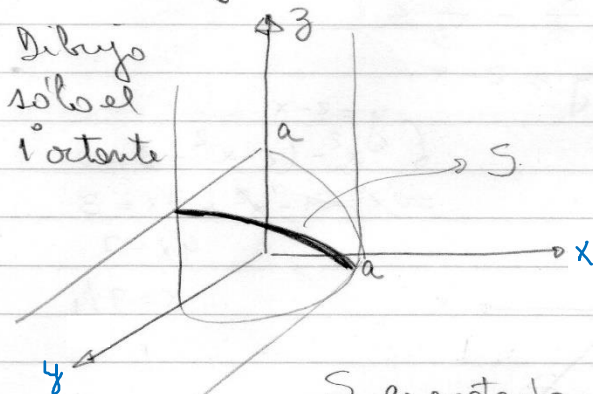
$$\text{Ext } S = \{(x, y) \in \mathbb{R}^2 \mid x^2 - y^2 > 3 \vee x + y < 2\}$$

$$S' = S \rightarrow S \text{ es cerrado}$$

S no es acotado. S no es compacto

S es conexo

3) d) $x^2 + y^2 = a^2$ $a > 0$ $x, y, z \geq 0$
 $x^2 + z^2 = a^2$



$$S^0 = \emptyset$$

$$\partial S = S$$

$$E_{\mathbb{R}^3} S = \mathbb{R}^3 - S$$

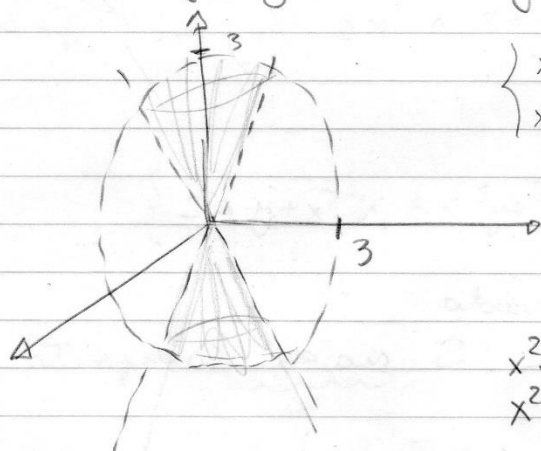
$$S' = S \rightarrow S \text{ es cerrado}$$

S es acotado $\rightarrow \exists E(0, 2500) \supset S$

S es compacto

S es conexo

3) e) $x^2 + y^2 < z^2$ $x^2 + y^2 + z^2 < 9$



$$\begin{cases} x^2 + y^2 = z^2 \\ x^2 + y^2 + z^2 = 9 \end{cases} \Rightarrow$$

$$\Rightarrow z^2 + z^2 = 9 \Rightarrow z^2 = 9/2$$

$$\Rightarrow z = \pm \frac{3}{\sqrt{2}}$$

$$x^2 + y^2 = z^2 \rightarrow \text{cono}$$

$$x^2 + y^2 + z^2 = 9 \rightarrow \text{esfera}$$

$$S^0 = S$$

$$\partial S = \left\{ (x, y, z) \in \mathbb{R}^3 / \left(x^2 + y^2 = z^2 \wedge -\frac{3}{\sqrt{2}} \leq z \leq \frac{3}{\sqrt{2}} \right) \vee \right. \\ \left. \vee \left(x^2 + y^2 + z^2 = 9 \wedge |z| \geq 3/\sqrt{2} \right) \right\}$$

$$E_{\mathbb{R}^3} S = \left\{ (x, y, z) \in \mathbb{R}^3 / x^2 + y^2 > z^2 \vee x^2 + y^2 + z^2 > 9 \right\}$$

$S^0 = S \rightarrow$ todos sus pts son int $\rightarrow S$ es abierto

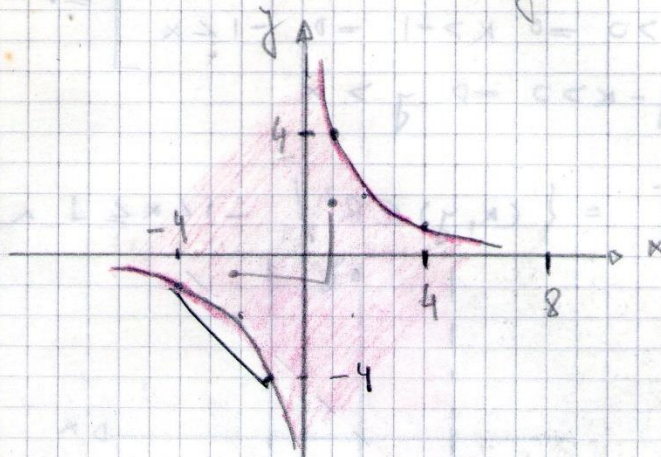
S acotado $\rightarrow \exists E(0, 2500) \supset S$

S no conexo $(0 \notin S!!)$

04) $xy \leq 4 \rightarrow \begin{cases} y \leq \frac{4}{x} & x > 0 \\ y \geq \frac{4}{x} & x < 0 \end{cases} \quad (\text{con } x \neq 0)$

Si $x=0 \rightarrow$ se verifica $xy \leq 4$

Si $y=0 \rightarrow$ se verifica $xy \leq 4$



Como $S = S \leq S \Rightarrow S$ es cerrado

S es convexo, ya que cualquier par de pts del conj pueden ser unidos mediante una poligonal incluida en el conj

Sin embargo no es convexo, ya que p.ej, los pts $(-4, -1)$ y $(-1, -4)$ se unen mediante un segmento pero este totalmente incluido en S

05)

a) $f(x,y) = \ln[(x+1)(y-2x)]$

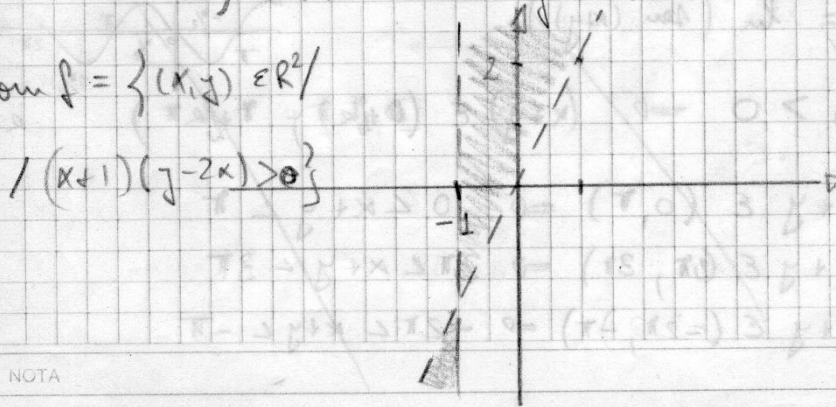
$(x+1)(y-2x) > 0$

i) $x+1 > 0 \wedge y-2x > 0 \Rightarrow x > -1 \wedge y > 2x$

ii) $x+1 < 0 \wedge y-2x < 0 \Rightarrow x < -1 \wedge y < 2x$

$\text{Dom } f = \{(x,y) \in \mathbb{R}^2 /$

$(x+1)(y-2x) > 0\}$



NOTA

5)b) $f_1 = \sqrt{1-x}$ $f_2 = \frac{1}{\sqrt{x+1}}$ $f_3 = \ln(y-x)$

$\text{Dom } \bar{f} = \text{Dom } f_1 \cap \text{Dom } f_2 \cap \text{Dom } f_3$

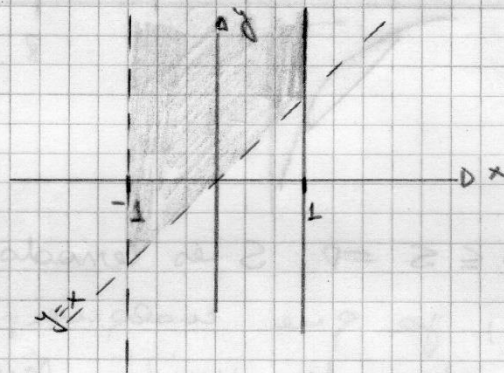
i) $1-x \geq 0 \Rightarrow 1 \geq x \Rightarrow x \leq 1$

ii) $x+1 > 0 \Rightarrow x > -1 \Rightarrow -1 < x$

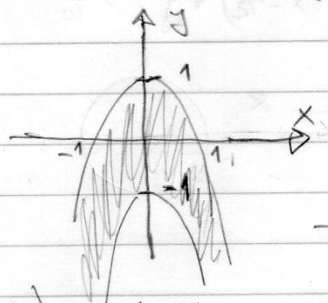
iii) $y-x > 0 \Rightarrow y > x$

$\Rightarrow -1 < x \leq 1$

$\text{Dom } \bar{f} = \{(x,y) \in \mathbb{R}^2 / -1 < x \leq 1 \wedge y > x\}$



$$5) c) 1 - (x^2 + y)^2 \geq 0 \Rightarrow (x^2 + y)^2 \leq 1 \Rightarrow$$



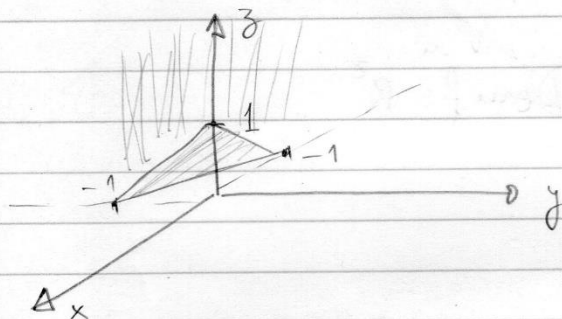
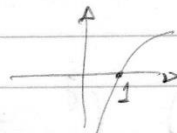
$$\Rightarrow |x^2 + y| \leq 1 \Rightarrow$$

$$-1 \leq x^2 + y \leq 1$$

$$-1 - x^2 \leq y \wedge y \leq 1 - x^2$$

$$d) \ln(3 - x - y) \geq 0 \Rightarrow$$

$$\Rightarrow 3 - x - y \geq 1 \Rightarrow 3 \geq 1 + x + y$$

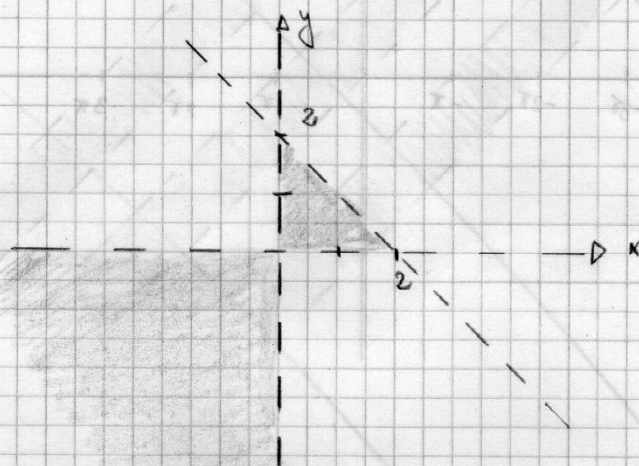


$$D = \{ (x, y, z) \in \mathbb{R}^3 / z \geq 1 + x + y \}$$

$$5) e) 2 - x - y > 0 \wedge xy > 0 \Rightarrow y < 2 - x \wedge xy > 0$$

$$\Rightarrow y < 2 - x \wedge [(x > 0 \wedge y > 0) \vee (x < 0 \wedge y < 0)]$$

$$\text{Dom } f = \{ (x, y) \in \mathbb{R}^2 / y < 2 - x \wedge xy > 0 \}$$

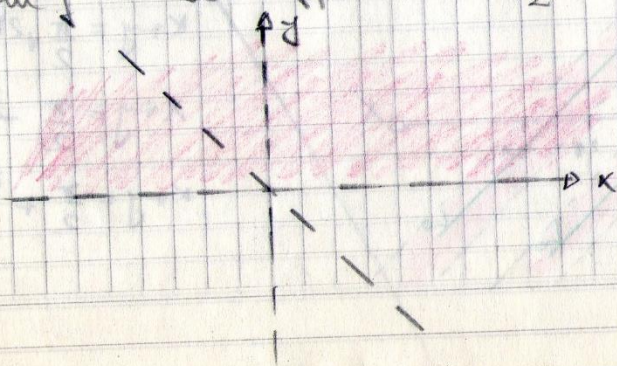


$$f) \quad f_1 = x^{-2} \quad f_2 = (x+y)^{-2} \sqrt{y}$$

$$i) \quad x \neq 0$$

$$ii) \quad x+y \neq 0 \wedge y > 0$$

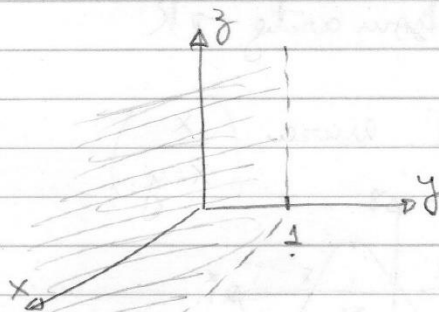
$$\text{Dom } \bar{f} = \text{Dom } f_1 \cap \text{Dom } f_2 = \{(x,y) \in \mathbb{R}^2 / x \neq 0 \wedge x+y \neq 0 \wedge y > 0\}$$



NOTA

5) g) $1-y > 0 \Rightarrow y < 1$

$\text{Dom } f = \{(x, y, z) \in \mathbb{R}^3 / y < 1\}$



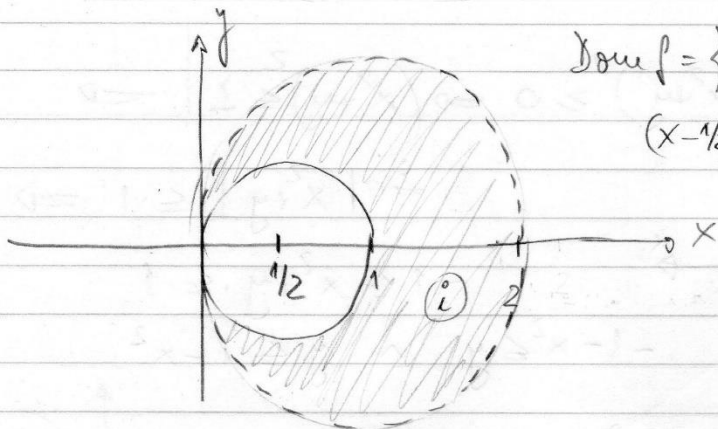
5 h) i) $x^2 + y^2 - x \geq 0 \wedge 2x - x^2 - y^2 > 0$

$(x^2 - x + \frac{1}{4}) + y^2 \geq \frac{1}{4} \wedge x^2 + y^2 - 2x < 0$

$(x - \frac{1}{2})^2 + y^2 \geq \frac{1}{4} \wedge x^2 - 2x + 1 + y^2 < 1$

ii) $x^2 + y^2 - x \leq 0 \wedge 2x - x^2 - y^2 < 0$

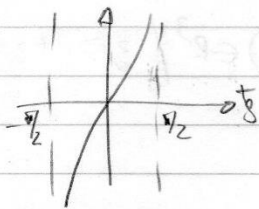
$(x - \frac{1}{2})^2 + y^2 \leq \frac{1}{4} \wedge (x-1)^2 + y^2 > 1 \rightarrow \emptyset$



$\text{Dom } f = \{(x, y) \in \mathbb{R}^2 /$

$(x - \frac{1}{2})^2 + y^2 \geq \frac{1}{4} \wedge (x-1)^2 + y^2 \geq 1\}$

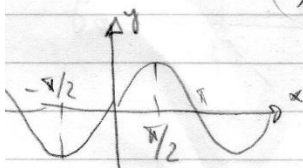
$$i) \int \frac{y}{x} \frac{1}{1+t^2} dt = \arctan t \Big|_x^y = \arctan y - \arctan x$$



$$\text{Dom } f = \mathbb{R}^2$$

Dom arctg $\rightarrow \mathbb{R}$

$$j) \text{ arctan} \left(\frac{x}{x+y} \right)$$



$$\text{arctan} : [-1, 1] \rightarrow [-\pi/2, \pi/2]$$

$$x+y \neq 0 \rightarrow y \neq -x$$

$$-1 \leq \frac{x}{x+y} \leq 1$$

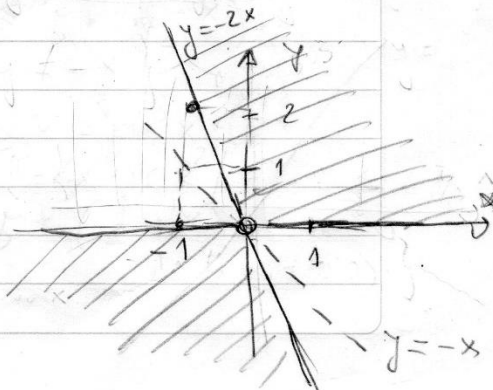
$$x+y > 0 \rightarrow -x-y \leq x \leq x+y$$

$$-2x \leq y \wedge y \geq 0$$

$$x+y < 0 \rightarrow -x-y \geq x \geq x+y$$

$$-2x \geq y \wedge y \leq 0$$

$$\text{Dom } f = \{(x, y) \in \mathbb{R}^2 / (y \geq -2x \wedge y \geq 0) \vee (y \leq -2x \wedge y \leq 0)\} - \{(0, 0)\}$$



Represente geoméricamente los conjuntos de nivel de los siguientes campos escalares: $f(x,y)=xy-2$

6)a) $f(x,y)=k \Rightarrow xy-2=k \Rightarrow xy=k+2$

$k=0 \rightarrow xy=2 \rightarrow y=\frac{2}{x}$

$k=-1 \rightarrow xy=1 \rightarrow y=\frac{1}{x}$

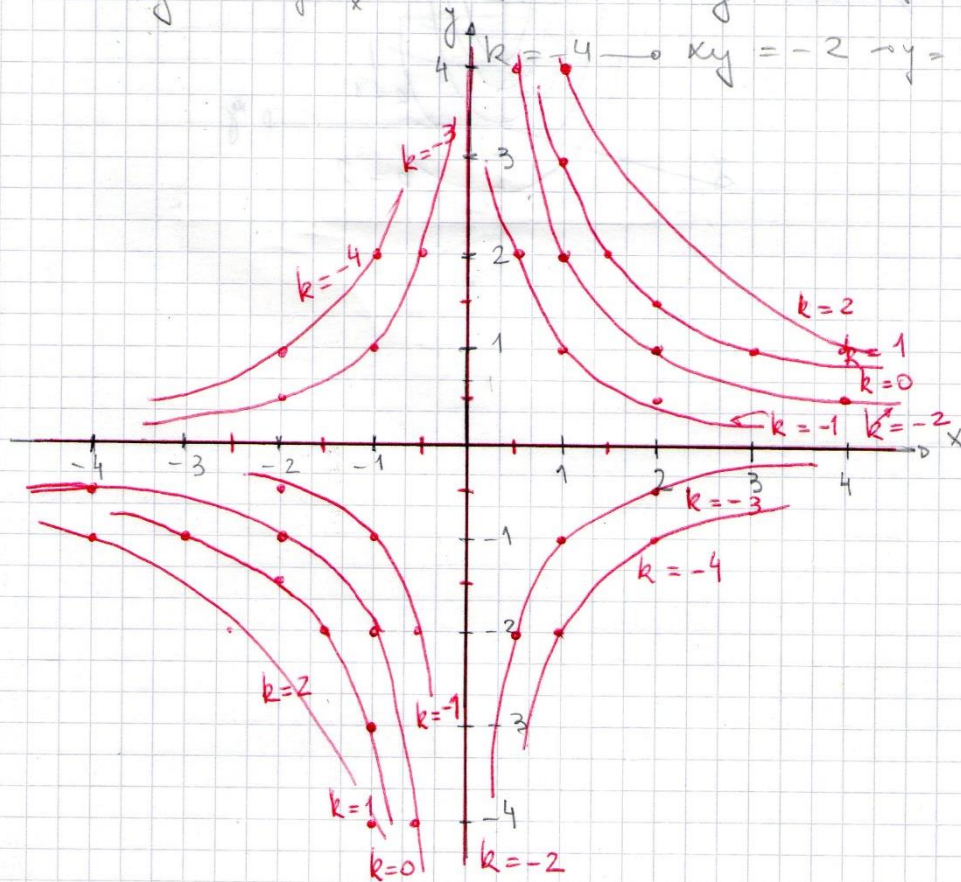
$k=1 \rightarrow xy=3 \rightarrow y=\frac{3}{x}$

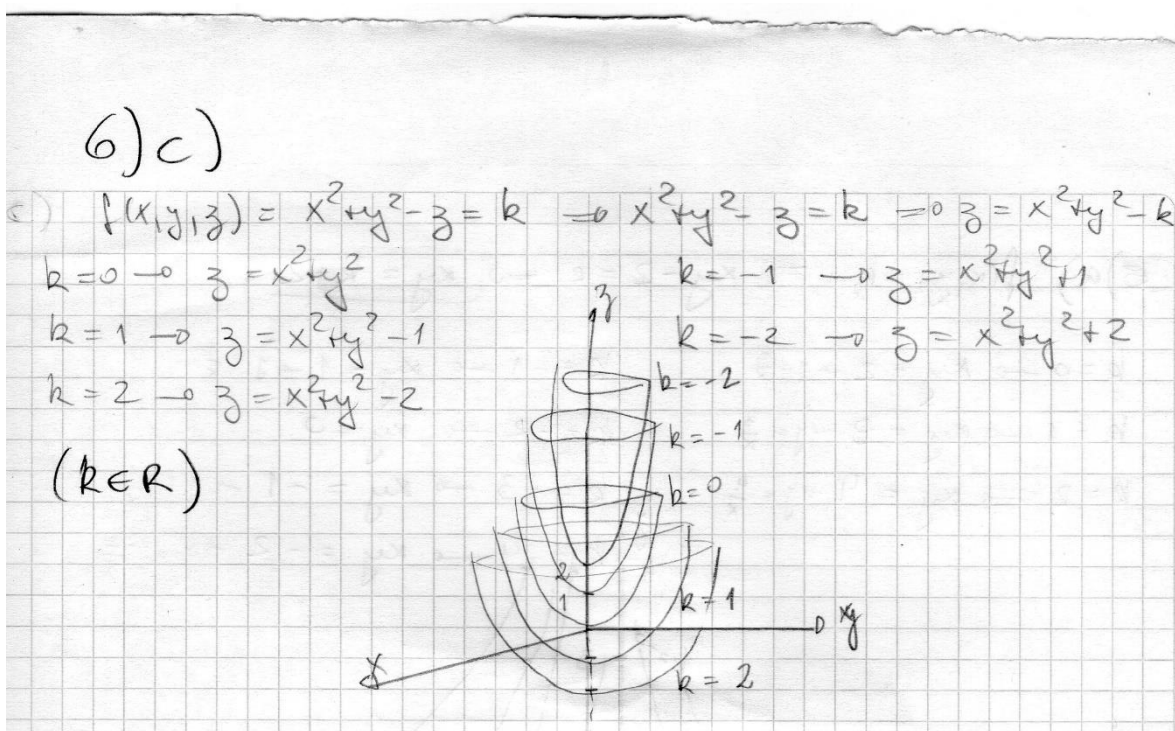
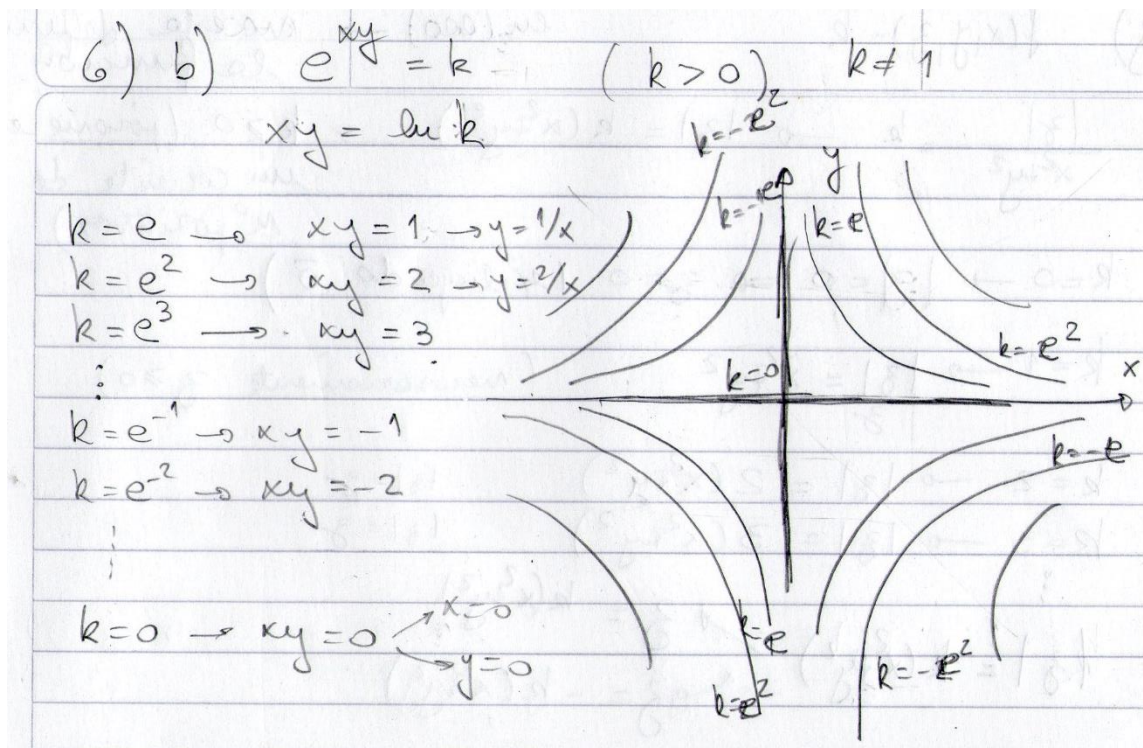
$k=-2 \rightarrow xy=0$

$k=2 \rightarrow xy=4 \rightarrow y=\frac{4}{x}$

$k=-3 \rightarrow xy=-1 \rightarrow y=-\frac{1}{x}$

$k=-4 \rightarrow xy=-2 \rightarrow y=-\frac{2}{x}$





6) d) $|x| + y = k \quad \wedge \quad k \in \mathbb{R}$

$$y = k - |x|$$

$$k=0 \rightarrow y = -|x|$$

$$k=1 \rightarrow y = 1 - |x|$$

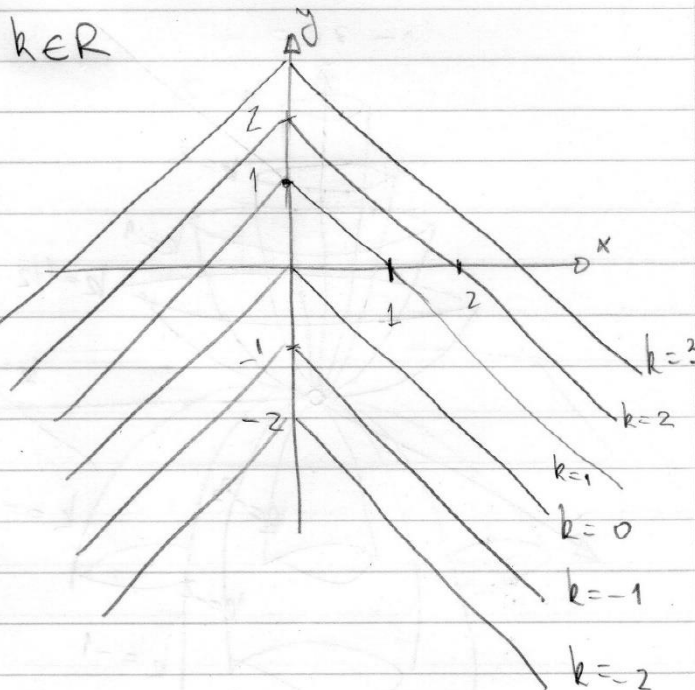
$$k=2 \rightarrow y = 2 - |x|$$

$\vdots$

$$k=-1 \rightarrow y = -1 - |x|$$

$$k=-2 \rightarrow y = -2 - |x|$$

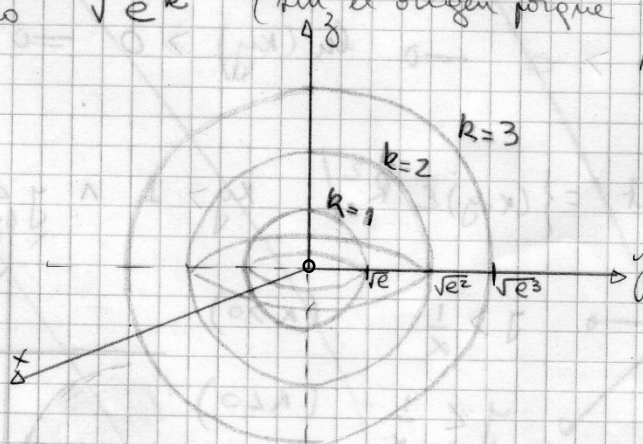
$\vdots$



Represente geométicamente los conjuntos de nivel de los siguientes campos escalares: $f(x,y,z) = \ln(x^2 + y^2 + z^2)$

6) e) $k = \ln(x^2 + y^2 + z^2) \Rightarrow e^k = x^2 + y^2 + z^2 \quad (k \in \mathbb{R})$

Cosméticos esféricos concéntricos, centro en el origen
y radio $\sqrt{e^k}$ (sin el origen porque $e^k \neq 0 \quad \forall k$) $\wedge$
 $\wedge \quad \vec{0} \notin \text{Dom} f$



$$6) f) \quad f(x, y) = k \Rightarrow k = \frac{x}{x^2 + y^2} \Rightarrow$$

$$\Rightarrow k(x^2 + y^2) = x$$

$$k=0 \rightarrow x=0$$

$$k \neq 0 \rightarrow x^2 + y^2 = \frac{x}{k} \Rightarrow x^2 - \frac{x}{k} + y^2 = 0 \Rightarrow$$

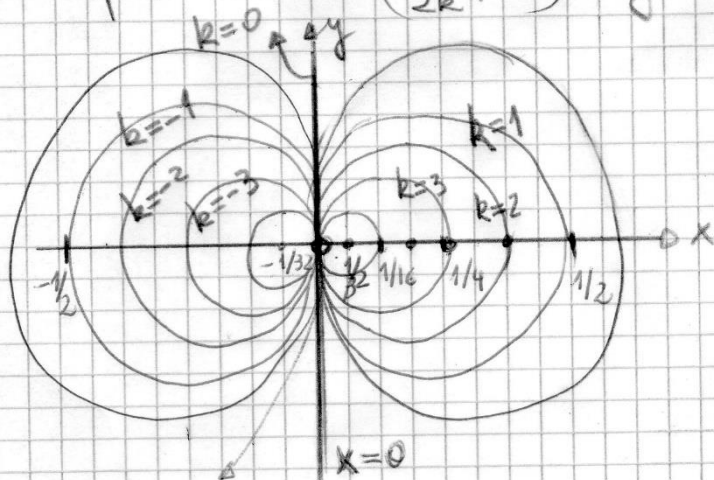
$$\Rightarrow x^2 - \frac{x}{k} + \frac{1}{4k^2} + y^2 = \frac{1}{4k^2} \Rightarrow \left(x - \frac{1}{2k}\right)^2 + y^2 = \left(\frac{1}{2k}\right)^2$$

$$k=1 \rightarrow \left(x - \frac{1}{2}\right)^2 + y^2 = \frac{1}{4}$$

$$k=2 \rightarrow \left(x - \frac{1}{4}\right)^2 + y^2 = \frac{1}{16}$$

$$k=-1 \rightarrow \left(x + \frac{1}{2}\right)^2 + y^2 = \frac{1}{4}$$

⋮
Circunf. de centro $\left(\frac{1}{2k}, 0\right)$ y radio $\left|\frac{1}{2k}\right|$ (con $k \neq 0$)



$(0,0) \notin \text{Dom } f$ (las curvas de nivel no se cortan, porque $0 \notin \text{Dom } f$)

$$k=1/2 \rightarrow (x^2 - 1)^2 + y^2 = 1$$

$$k=-1/2 \rightarrow (x+1)^2 + y^2 = 1$$

$$x \neq 0 \wedge y \neq 0$$

$$\text{Dom } f = \{x \in \mathbb{R}^3 / x^2 + y^2 \neq 0\}$$

$$g) f(x, y, z) = k$$

en el eje z

→ no está definida la función

$$\frac{|z|}{x^2 + y^2} = k \Rightarrow |z| = k(x^2 + y^2)$$

$k \geq 0$ (porque es un cociente de n° positivos)

$$k=0 \rightarrow |z|=0 \Rightarrow z=0 \text{ (excluyendo } \bar{0} \text{)}$$

$$k=1 \rightarrow |z| = x^2 + y^2$$

$$\begin{cases} \text{Si } z \geq 0 : z = x^2 + y^2 \\ \text{Si } z < 0 : -z = x^2 + y^2 \Rightarrow z = -(x^2 + y^2) \end{cases}$$

$$k=2 \rightarrow |z| = 2(x^2 + y^2)$$

$$k=3 \rightarrow |z| = 3(x^2 + y^2)$$

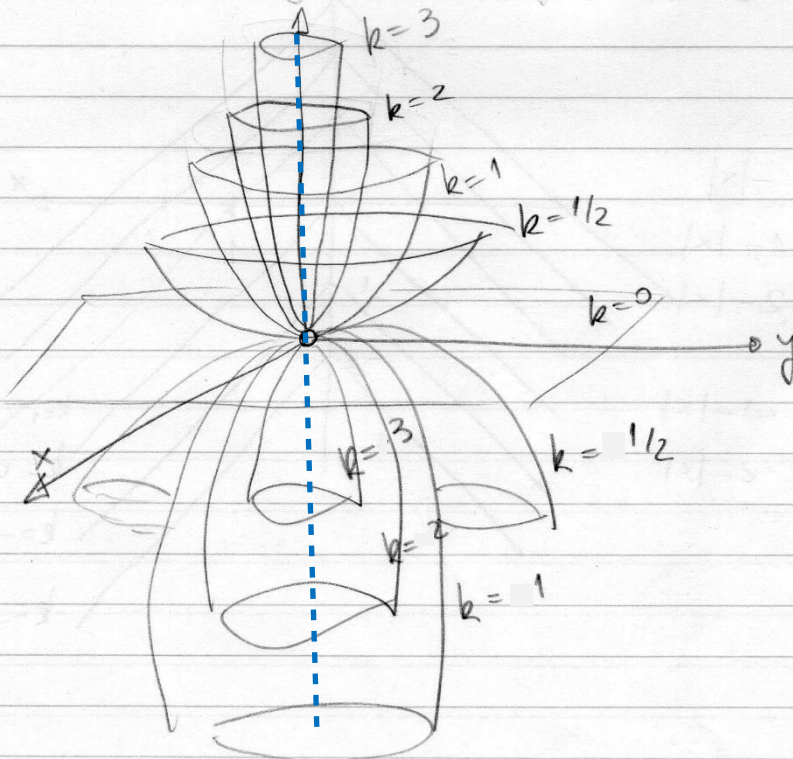
⋮

$$|z| = k(x^2 + y^2) \rightarrow z = k(x^2 + y^2) \quad \text{si } z > 0$$

$$|z| = k(x^2 + y^2) \rightarrow z = -k(x^2 + y^2) \quad \text{si } z < 0$$

$$k=-2 \rightarrow |z| = -2(x^2 + y^2)$$

$$-z = 2(x^2 + y^2) \Rightarrow z = -2(x^2 + y^2)$$



7) a) $z = x^2 + y^2$ paraboloida circular

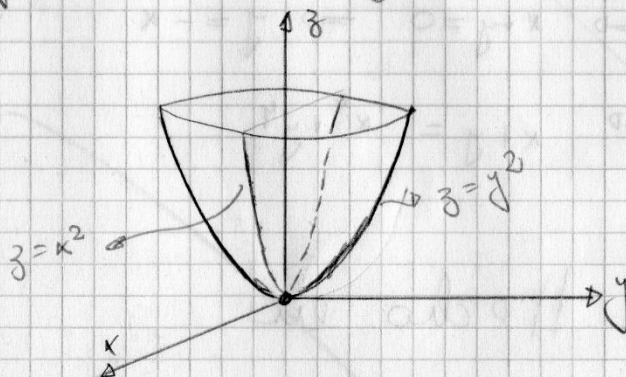
intersec $\pi(xy) \rightarrow z=0 \Rightarrow x^2 + y^2 = 0 \Rightarrow (x,y) = (0,0)$

intersec $\pi(xz) \rightarrow y=0 \Rightarrow z = x^2$ (parabola)

intersec $\pi(yz) \rightarrow x=0 \Rightarrow z = y^2$ (parabola)

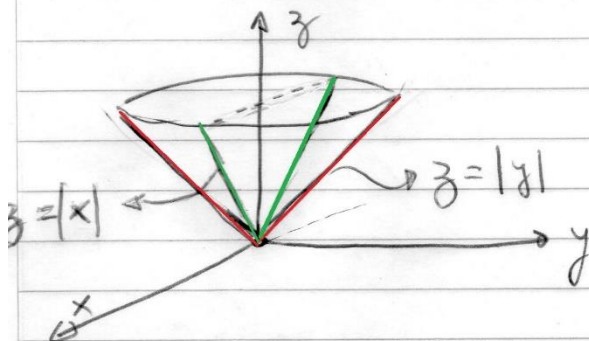
$$\text{Im } f = \mathbb{R}_0^+$$

Postiv. $f = \mathbb{R}^2 - \{(0,0)\}$



7) b) $z = \sqrt{x^2 + y^2}$ ($\rightarrow z^2 = x^2 + y^2$ doble cono a 45°)

superficie conica con $z \geq 0$



$$\text{Dom } f = \mathbb{R}_+^0$$

$$\text{Dom } f = \mathbb{R}^2$$

intersecc pl (xy) $\rightarrow z=0$

$$\left. \begin{array}{l} z=0 \\ z=\sqrt{x^2+y^2} \end{array} \right\} \Rightarrow \left. \begin{array}{l} z=0 \\ x^2+y^2=0 \end{array} \right\} \Rightarrow (x,y,z) = (0,0,0)$$

intersecc pl (xz) $\rightarrow y=0$

$$\left. \begin{array}{l} y=0 \\ z=\sqrt{x^2+y^2} \end{array} \right\} \Rightarrow \left. \begin{array}{l} y=0 \\ z=\sqrt{x^2} \end{array} \right\} \Rightarrow \left. \begin{array}{l} y=0 \\ z=|x| \end{array} \right\}$$

intersecc pl (yz) $\rightarrow x=0$

$$\left. \begin{array}{l} x=0 \\ z=\sqrt{x^2+y^2} \end{array} \right\} \Rightarrow \left. \begin{array}{l} x=0 \\ z=\sqrt{y^2} \end{array} \right\} \Rightarrow \left. \begin{array}{l} x=0 \\ z=|y| \end{array} \right\}$$

Positividad $\rightarrow f(x,y) > 0$

$$\sqrt{x^2+y^2} > 0 \Leftrightarrow (x,y) \neq (0,0)$$

$$C^+ = \mathbb{R}^2 - \{(0,0)\}$$

$$c) \quad z = \sqrt{9 - x^2 - y^2} \Rightarrow z^2 = 9 - x^2 - y^2 \Rightarrow x^2 + y^2 + z^2 = 9$$

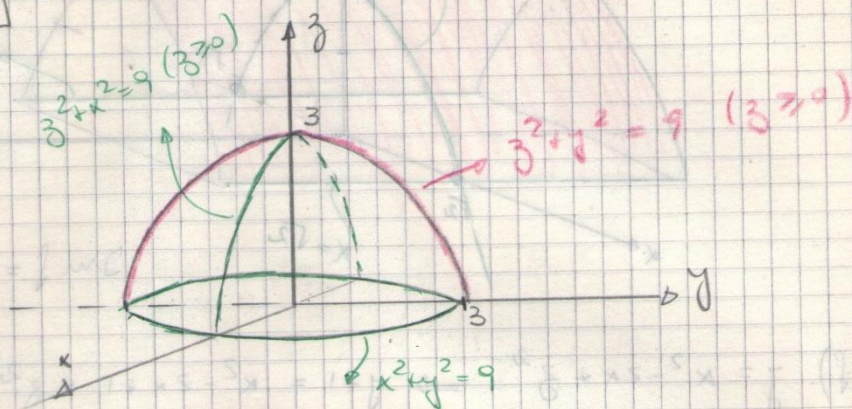
semi esfero superior, centro en el origen y radio 3

$$\text{intersec } p(x, y) \rightarrow z = 0 \Rightarrow \sqrt{9 - x^2 - y^2} = 0 \Rightarrow x^2 + y^2 = 9$$

$$\text{intersec } p(x, z) \rightarrow y = 0 \Rightarrow z = \sqrt{9 - x^2} \Rightarrow z^2 = 9 - x^2 \Rightarrow x^2 + z^2 = 9 \quad (\text{con } z \geq 0)$$

$$\text{intersec } p(y, z) \rightarrow x = 0 \Rightarrow z = \sqrt{9 - y^2} \Rightarrow z^2 = 9 - y^2 \Rightarrow z^2 + y^2 = 9 \quad (\text{con } z \geq 0)$$

$$w_f = [0, 3]$$



$$c) \quad f(x, y) = \sqrt{9 - x^2 - y^2}$$

$$9 - x^2 - y^2 \geq 0 \Rightarrow 9 \geq x^2 + y^2$$

$$\text{posit } f = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 9\}$$



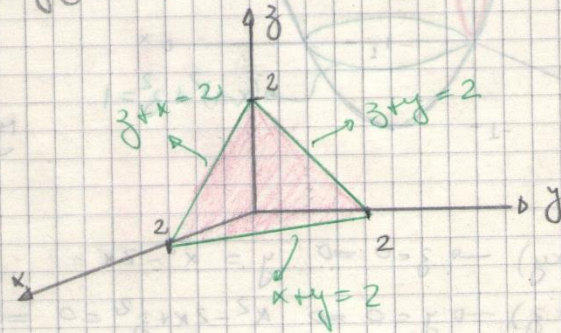
7) d) $z = 2 - x - y = 0 \Rightarrow z + x + y = 2$ plano

intersec $pl(xy) \rightarrow z = 0 \Rightarrow x + y = 2$

intersec $pl(xz) \rightarrow y = 0 \Rightarrow z + x = 2$

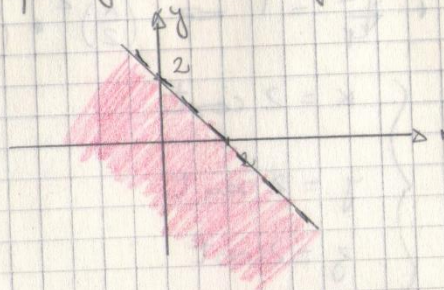
intersec $pl(yz) \rightarrow x = 0 \Rightarrow z + y = 2$

$\text{Im } f = R$



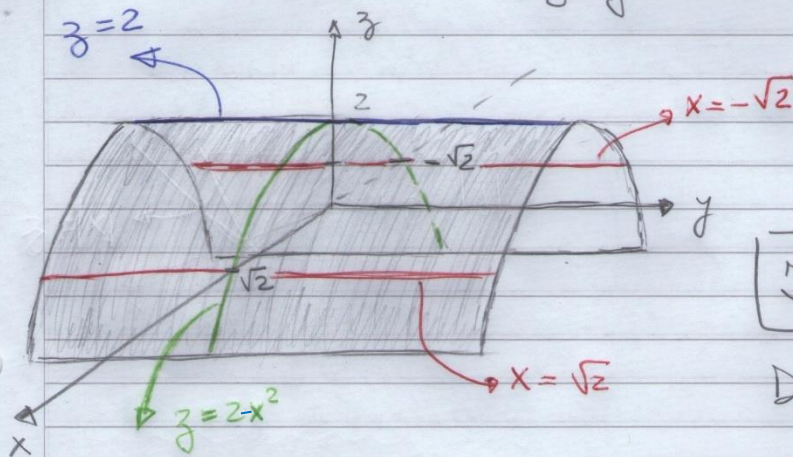
d) $f(x, y) = 2 - x - y > 0 \Rightarrow 2 - x > y$

port $f = \{(x, y) \in R^2 \mid y < 2 - x\}$



7) e) $z = 2 - x^2$

superficie cilíndrica de eje y



$$\text{Im } f = (-\infty, 2]$$

$$\text{Dom } f = \mathbb{R}^2$$

intersecc pl(xy) $\rightarrow z = 0$

$$\left. \begin{array}{l} z = 0 \\ z = 2 - x^2 \end{array} \right\} = 0 \quad \left. \begin{array}{l} z = 0 \\ 2 - x^2 = 0 \end{array} \right\} = 0 \quad \left. \begin{array}{l} z = 0 \\ x^2 = 2 \end{array} \right\} = 0 \quad \left. \begin{array}{l} z = 0 \\ x = \pm \sqrt{2} \end{array} \right\}$$

dos rectas
en el pl(xy)

intersecc pl(yz) $\rightarrow x = 0$

$$\left. \begin{array}{l} x = 0 \\ z = 2 - x^2 \end{array} \right\} = 0 \quad \left. \begin{array}{l} x = 0 \\ z = 2 \end{array} \right\} \quad \text{recta en el pl(yz)}$$

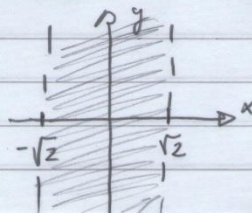
intersecc pl(xz) $\rightarrow y = 0$

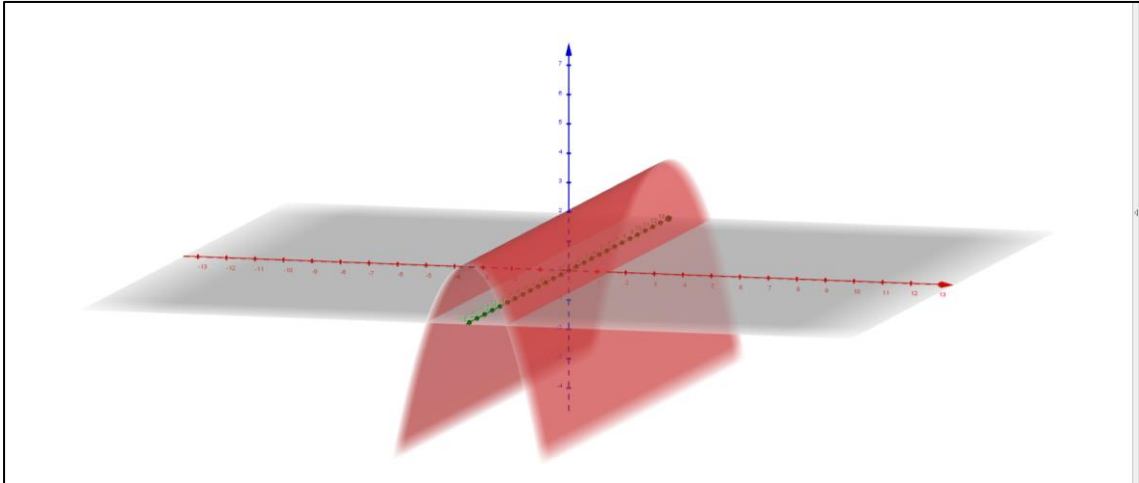
$$\left\{ \begin{array}{l} y = 0 \\ z = 2 - x^2 \end{array} \right. \quad \text{parábola en el pl(xz)}$$

Positividad $\rightarrow f(x,y) > 0 \Rightarrow 2 - x^2 > 0 \Rightarrow$

$$\Rightarrow x^2 < 2 \Rightarrow |x| < \sqrt{2} \Rightarrow -\sqrt{2} < x < \sqrt{2}$$

$$C^+ = \{(x,y) \in \mathbb{R}^2 / |x| < \sqrt{2}\}$$





Gráfica del campo escalar $f(x,y) = 2 - x^2$ (Ecuación de la gráfica $z = 2 - x^2$)

Para cada uno de los siguientes campos escalares definidos en su dominio natural: determine el conjunto imagen, halle el conjunto de positividad, represente la gráfica en el espacio xyz y analice las intersecciones con los planos coordenados

7) f) $f(x, z) = x^2 - 2x + z^2$

En este caso es $y = f(x, z)$ $\therefore$ graf $f \rightarrow y = f(x, z)$

Completamos cuadrados

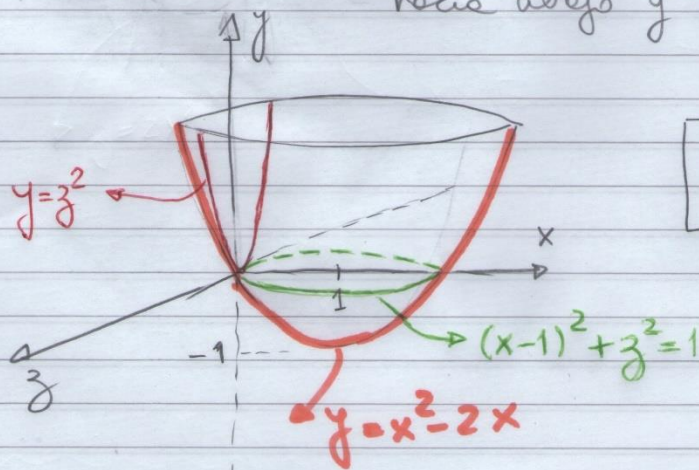
$$y = x^2 - 2x + z^2 = 0 \quad y + 1 = x^2 - 2x + 1 + z^2 = 0$$

$$\Rightarrow y = (x-1)^2 + z^2 - 1$$

Paraboloides circular
de eje y, desplazado
hacia abajo y hacia la derecha

$$\text{Dom } f = \mathbb{R}^2$$

$$\text{Im } f = [-1, +\infty)$$



intersecc. pl(xy) $\rightarrow z=0$

$$\left. \begin{array}{l} z=0 \\ y = x^2 - 2x + z^2 \end{array} \right\} \Rightarrow \left. \begin{array}{l} z=0 \\ y = x^2 - 2x \end{array} \right\}$$

parábola en el pl(xy)

intersecc. pl(xz) $\rightarrow y=0$

$$\left. \begin{array}{l} y=0 \\ y = x^2 - 2x + z^2 \end{array} \right\} \Rightarrow \left. \begin{array}{l} y=0 \\ x^2 - 2x + z^2 = 0 \end{array} \right\}$$

$$\Rightarrow \left. \begin{array}{l} y=0 \\ x^2 - 2x + 1 + z^2 = 1 \end{array} \right\} \Rightarrow \left. \begin{array}{l} y=0 \\ (x-1)^2 + z^2 = 1 \end{array} \right\} \text{círculo en el pl } y=0$$

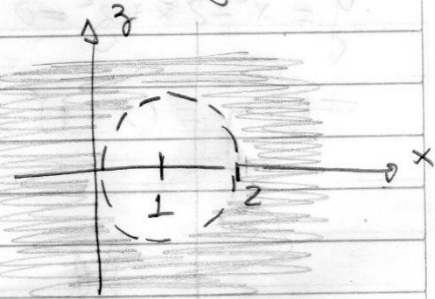
intersecc pl (yz) $\rightarrow x=0$

$$\left. \begin{array}{l} x=0 \\ y=x^2-2x+z^2 \end{array} \right\} \Rightarrow \left. \begin{array}{l} x=0 \\ y=z^2 \end{array} \right\} \text{ parabola en el pl } x=0$$

Positividad $f(x,z) > 0 \Rightarrow x^2-2x+z^2 > 0 \Rightarrow$

$$\Rightarrow (x-1)^2 + z^2 > 1$$

$$C^+ = \left\{ (x,y) \in \mathbb{R}^2 / (x-1)^2 + z^2 > 1 \right\}$$



$$8) a) f(x, y) = \frac{1}{\sqrt{x^2 + y^2 - 1}}$$

$$x^2 + y^2 - 1 > 0 \Rightarrow x^2 + y^2 > 1$$

$$b) x^2 + y^2 + 2x \leq 8 \Rightarrow x^2 + 2x + 1 + y^2 \leq 8 + 1 \Rightarrow (x+1)^2 + y^2 \leq 9 \Rightarrow 9 - (x+1)^2 - y^2 \geq 0$$

$$f(x, y) = \sqrt{9 - (x+1)^2 - y^2}$$

$$c) -1 \leq x+y < 3$$

$$-1 \leq x+y \quad \wedge \quad x+y < 3$$

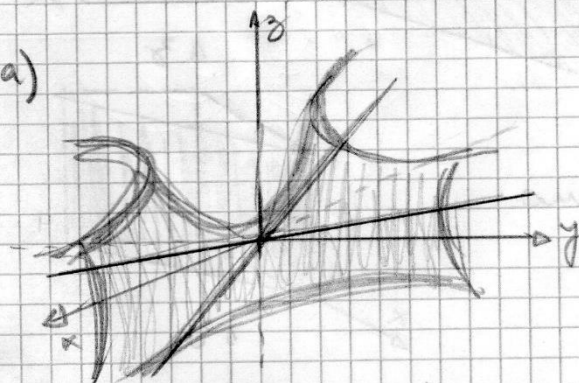
$$0 \leq x+y+1 \quad \wedge \quad 0 < 3-x-y$$

$$f(x, y) = \frac{\sqrt{x+y+1}}{\sqrt{3-x-y}}$$

$$d) f(x, y) = \frac{1}{\sqrt{(x-1)^2 + (y-2)^2}}$$

9)

a)

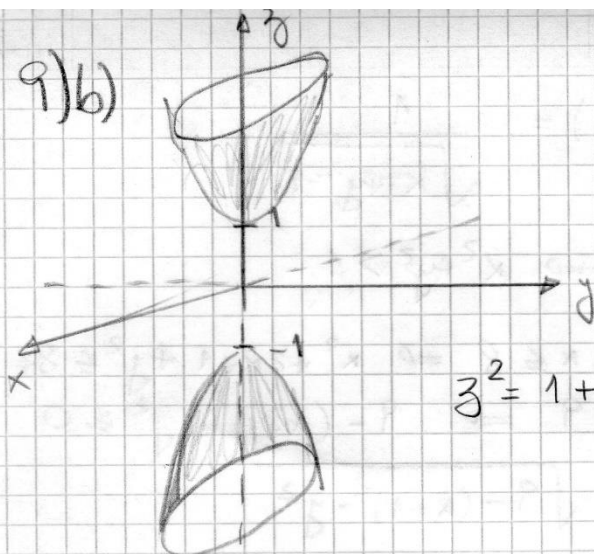


siège de montar

paraboloïde hyperbolique

$$z=0 \Rightarrow \begin{cases} x=2y \Rightarrow y=x/2 \\ x=-2y \Rightarrow y=-x/2 \end{cases}$$

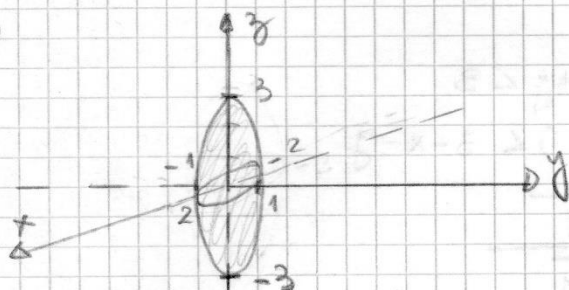
9)b)



hiperboloide
de 2 folhas
elíptico

$$z^2 = 1 + x^2 + 4y^2$$

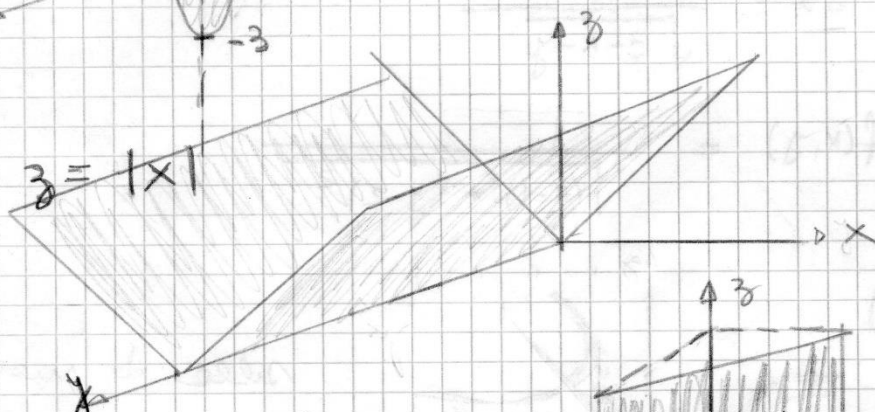
9)c)



Elipsoide

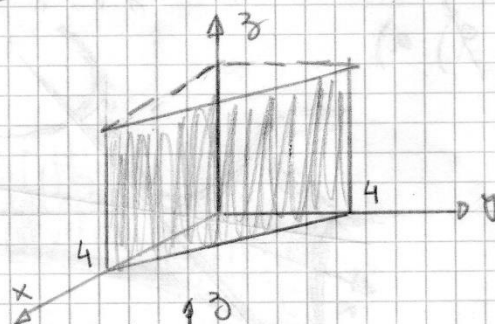
9)d)

$$z = |x|$$



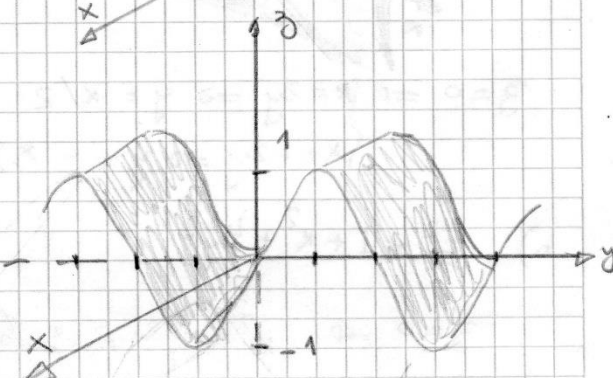
9)e)

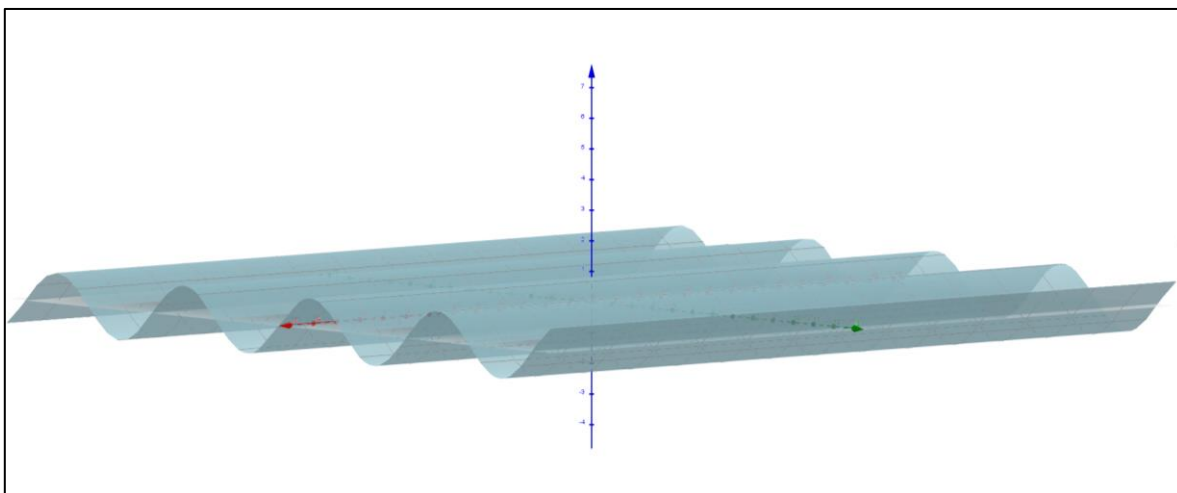
$$x + y = 4 \rightarrow \text{plano}$$



9)f)

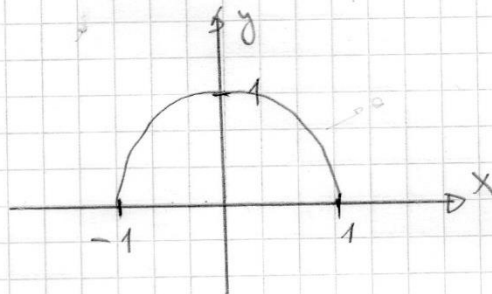
$z = \sin y$
superfície cilíndrica de eixo x
"chapa
onduleada"





Gráfica del conjunto de puntos $z = \sin(y)$

10) a) $\vec{g}(u) = (\cos u, \sin u)$ $u \in [0, \pi]$
 $\begin{cases} x = \cos u \\ y = \sin u \end{cases} \Rightarrow x^2 + y^2 = 1 \quad \downarrow 1^\circ y \geq 0 \text{ mod}$

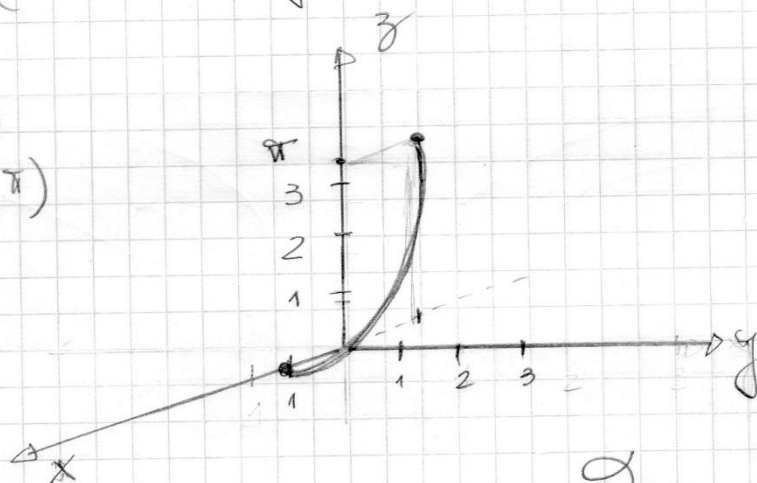


b) $\vec{g}(u) = (\cos u, \sin u, u)$ $u \in [0, \pi]$

$\begin{cases} x = \cos u \\ y = \sin u \end{cases} \Rightarrow x^2 + y^2 = 1 \quad z = u \quad \uparrow$

$\vec{g}(0) = (1, 0, 0)$

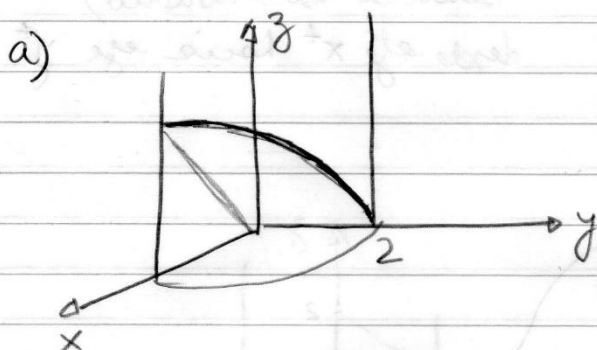
$\vec{g}(\pi) = (-1, 0, \pi)$



Li $u \in \mathbb{R}$ es um "resorte"



$$11) \begin{cases} x^2 + y^2 = 4 \\ z = x \end{cases} \quad 1^\circ \text{octante}$$



$x^2 + y^2 = 4$ cilindro
circular
recto

$z = x$ plano

$$b) \quad x^2 + y^2 = 4 \Rightarrow \frac{x^2}{4} + \frac{y^2}{4} = 1 \Rightarrow \left(\frac{x}{2}\right)^2 + \left(\frac{y}{2}\right)^2 = 1$$

$$\begin{cases} \frac{x}{2} = \cos t \Rightarrow x = 2 \cos t \\ \frac{y}{2} = \sin t \Rightarrow y = 2 \sin t \end{cases}$$

$$\begin{cases} \frac{x}{2} = \cos t \Rightarrow x = 2 \cos t \\ \frac{y}{2} = \sin t \Rightarrow y = 2 \sin t \end{cases}$$

Siendo $z = x$ resulte $z = 2 \cos t$

Como la curva está ubicada sólo en el
1º octante, "barrerá" un ángulo de $\pi/2$

(La curva completa "barrerá" un ángulo
de 2π , media curva uno de π y así sucesi-
vamente)

Luego: $\vec{f}: [0, \pi/2] \rightarrow \mathbb{R}^3 / \vec{f}(t) = (2 \cos t, 2 \sin t, 2 \cos t)$

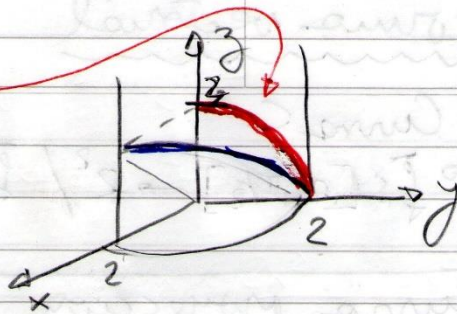
Podemos analizar el sentido de circulación:

Pto inicial $\rightarrow \vec{f}(0) = (2, 0, 2)$

Punto final $\rightarrow \vec{f}(\pi/2) = (0, 2, 0)$

Proyección plano (yz)

Ecuación $p(yz): x=0$



Forma vectorial

Curve C

$$\vec{r}: [0, \pi/2] \rightarrow \mathbb{R}^3 / \vec{r}(t) = (2\cos t, 2\sin t, 2\cos t)$$

= 0 proyección (yz)

Curve proyección

$$\bar{h}: [0, \pi/2] \rightarrow \mathbb{R}^3 / \bar{h}(t) = (0, 2\sin t, 2\cos t)$$

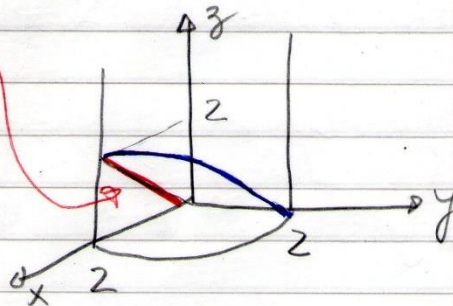
Forma cartesiana

$$\left. \begin{array}{l} x=0 \\ y=2\sin t \\ z=2\cos t \end{array} \right\} \begin{array}{l} x=0 \\ y^2 + z^2 = 4 \end{array}$$

$$\text{Luego: } \left\{ \begin{array}{l} y^2 + z^2 = 4 \\ x=0 \end{array} \right.$$

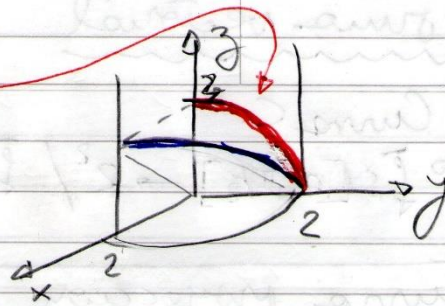
Proyección plano (xz)

Ecuación $p(xz): y=0$



Proyección plano (yz)

Ecuación $\pi(yz)$: $x=0$



Forma vectorial

Curva C

$$\vec{f}: [0, \pi/2] \rightarrow \mathbb{R}^3 / \vec{f}(t) = (2\cos t, 2\sin t, 2\cos t)$$

$= 0$ proyección (yz)

Curva proyección

$$\bar{h}: [0, \pi/2] \rightarrow \mathbb{R}^3 / \bar{h}(t) = (0, 2\sin t, 2\cos t)$$

Forma cartesiana

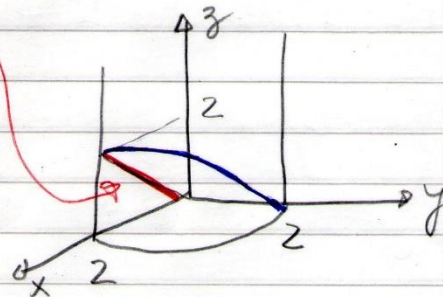
$$\left. \begin{array}{l} x=0 \\ y=2\sin t \\ z=2\cos t \end{array} \right\} \begin{array}{l} x=0 \\ y^2 + z^2 = 4 \end{array}$$

Luego:

$$\left\{ \begin{array}{l} y^2 + z^2 = 4 \\ x=0 \\ 1^\circ \text{ octante} \end{array} \right.$$

Proyección plano (xz)

Ecuación $\pi(xz)$: $y=0$



Forma vectorial

Curve C

$$\vec{f}: [0, \pi/2] \rightarrow \mathbb{R}^3 / \vec{f}(t) = (2\cos t, 2\sin t, 2\cos t)$$

= 0 proyección (x3)

Curve proyección

$$\bar{m}: [0, \pi/2] \rightarrow \mathbb{R}^3 / \bar{m}(t) = (2\cos t, 0, 2\cos t)$$

Forma cartesiana

$$\begin{cases} x = 2\cos t \\ y = 0 \\ z = 2\cos t \end{cases} \rightarrow \begin{cases} x = z \\ y = 0 \end{cases}$$

Luego

$$\begin{cases} x = z \\ y = 0 \\ \text{con } 0 \leq x \leq 2 \end{cases}$$

Se agrega la condición $0 \leq x \leq 2$ para no indicar toda la recta, sino solamente el segmento proyección